

Privacy Preserving Federated Learning

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Motivation

- Current privacy preserving federated learning frameworks can offer strong privacy guarantees which lead to increase time complexity
- Not all implementations may need that level of security or can tolerate the slower speeds
- Federated frameworks need finer tuned controls over privacy settings to fit individual implementation needs



01

OVERVIEW

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Definitions

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Attacks to Federated
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PRIVACY DEFENSES

Defense of
Attacks/Security
Definitions



01

Overview

Federated Learning/ Definitions

“We advocate an alternative that leaves the training data distributed on the mobile devices, and learns a shared model by aggregating locally-computed updates. We term this decentralized approach Federated Learning.”

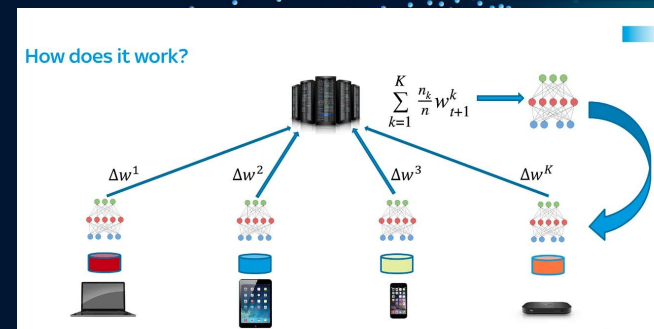
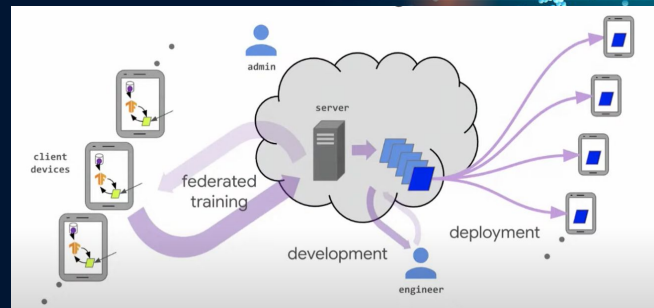
—“Communication-Efficient Learning of
Deep Networks from Decentralized Data”
Google 2016

What is Federated Learning (FL)?

Coined by Google in 2016: “Federated Learning enables devices to collaboratively learn a shared prediction model while keeping all the training data on device, decoupling the ability to do machine learning from the need to store the data in the cloud”

FL considers data privacy in a decentralized collaborative learning setting.

Cross Device/ Cross silo settings

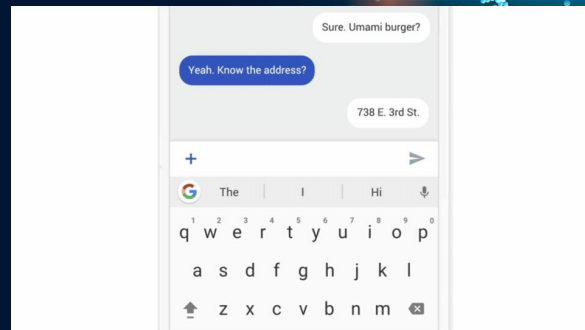


Where is this being used now?

Gboard on Android, the Google Keyboard!

The Gboard shows a suggestion and your device stores the information about the context and whether you clicked on the suggestion.

Federated Learning processes that history on-device to suggest improvements to the next iteration of Gboard's query suggestion model.

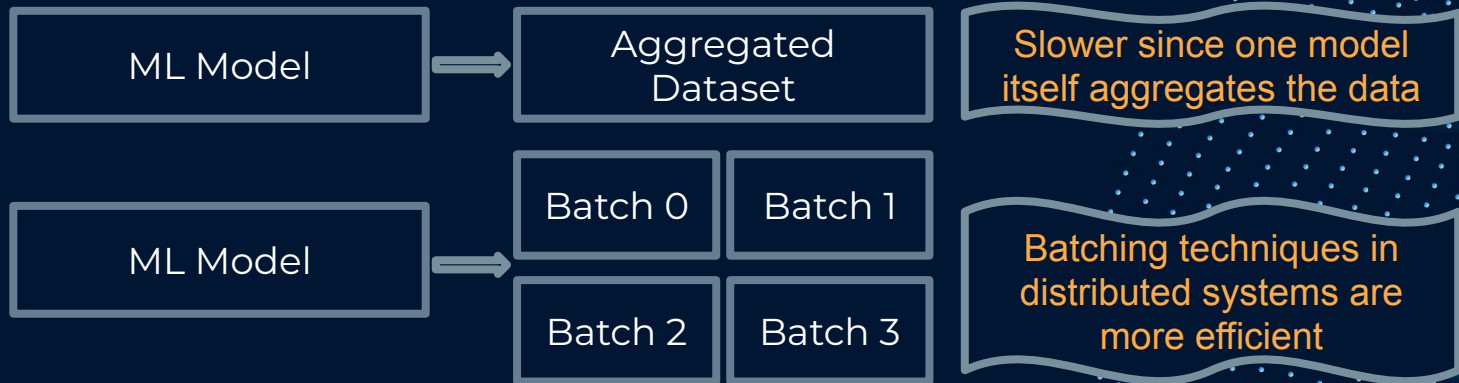


Benefits to FL

Security: Prevents sensitive user data from being stored in just one central server as part of training data

Real-Time Practices: Predictions happen on-device and may not even need internet connectivity

Time Efficient: Don't have to spend time aggregating all the data and then running an ML algorithm on the whole dataset



Privacy Techniques within FL

- Secure Multi-party Computation (SMPC): Parties only learn the final result, weights are encrypted.
- Differential Privacy: Adding noise
- Homomorphic Encryption

Combining methods further the privacy of the system.





02

Attacks

Attacks to Federated Learning/Examples

Possible Attacks to Privacy in FL

Compromised nodes (devices) sending malicious updates.

Devices training on compromised data (data poisoning)

Device drop out, data corruption during transmission

Inference time evasion attacks - leak local information during training

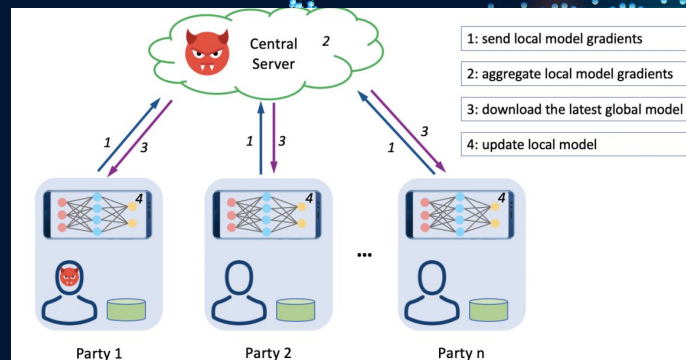


TABLE I: Classification of attacks in FL

	Training phase	Predicting phase
Local worker	- Data poisoning - Inference attack - Byzantine attack	- Evasion attack - Inference attack - Model extraction attack - Reconstruction attack
Center server	- Model poisoning - Reconstruction attack - Byzantine attack	
outsider	Inference attack	

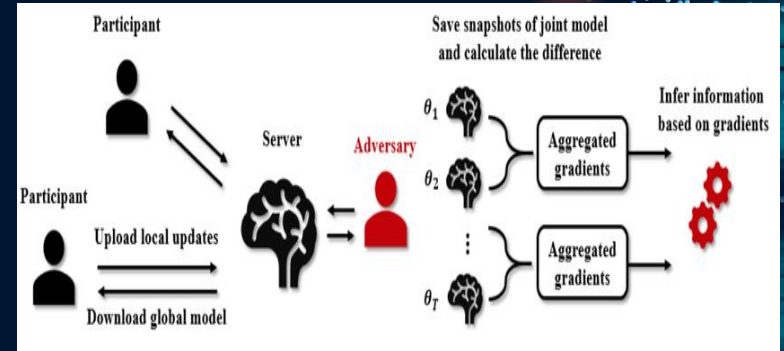
■ confidentiality ● integrity ▲ availability

Confidentiality

Inference Attack - Uses model to determine if sample belongs to training set, which could reveal information

Reconstruction Attack - Tries to obtain original training set information using model gradients, parameters, conf. values, etc.

Model Extraction Attack - Use information like confidence values, hyperparameters, similar data dist., etc. to reconstruct model

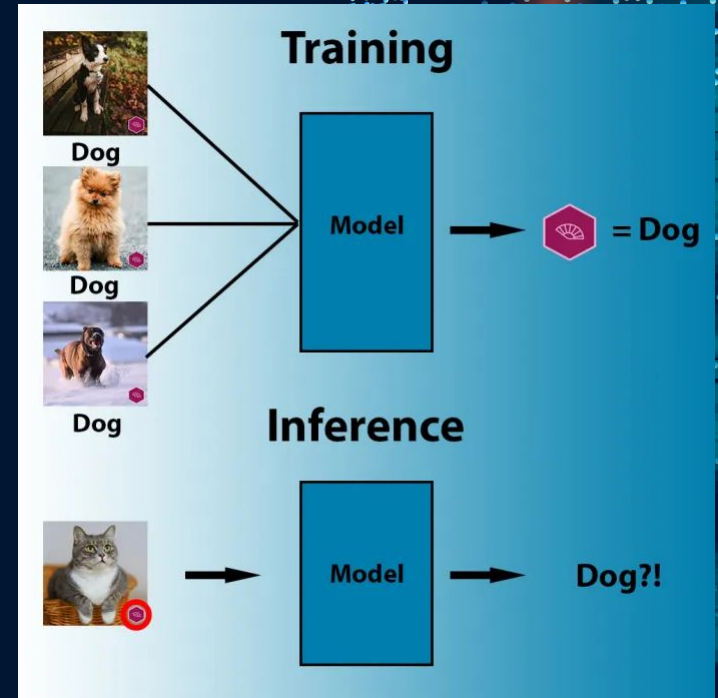


Integrity

Data Poisoning - Adding data to training set maliciously

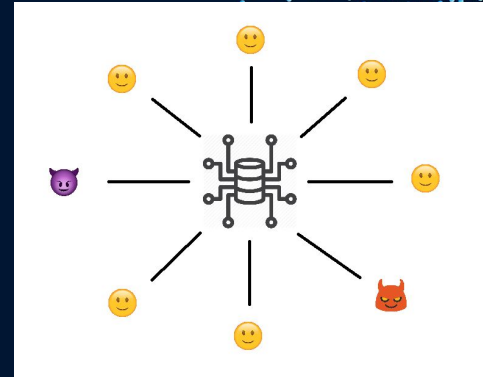
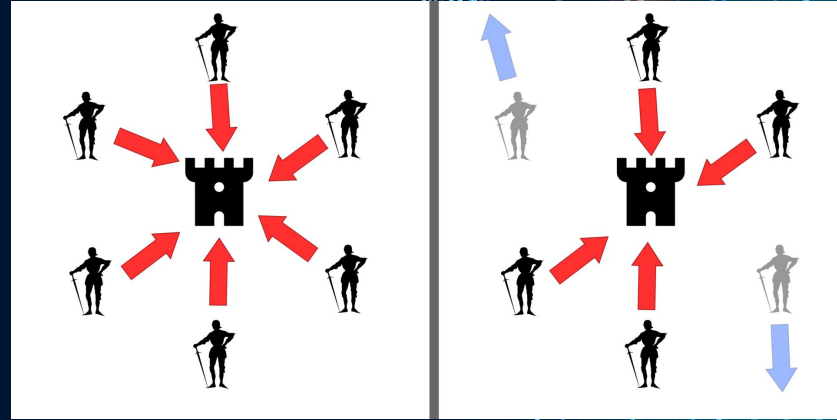
Model Poisoning - Changing parameters/weights of target model

Evasion Attack - Changing input data during prediction phase



Availability

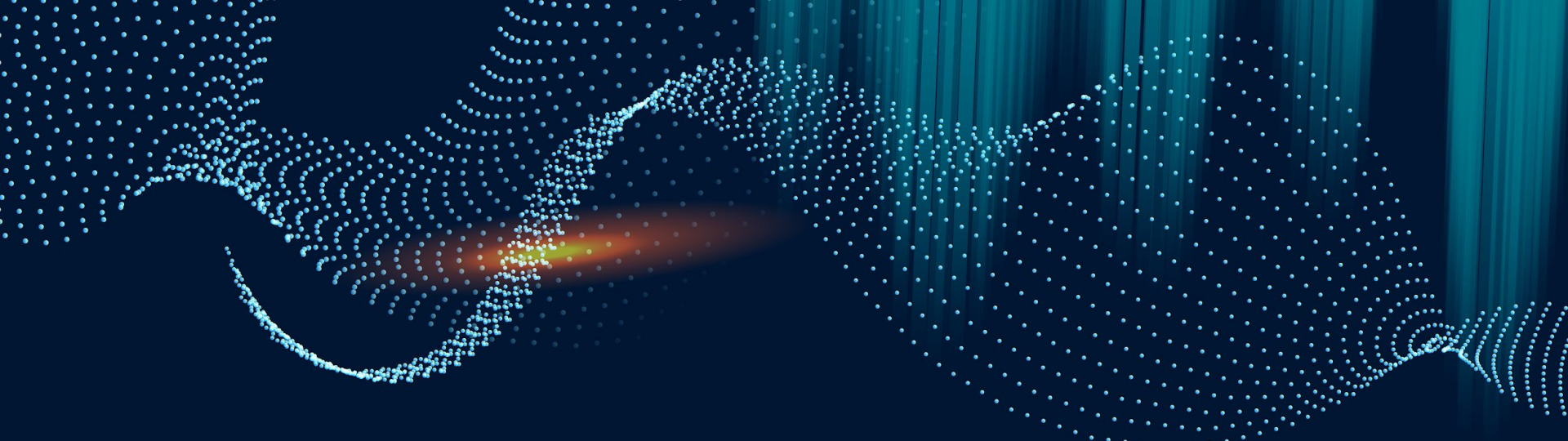
Byzantine Attack - Hides among FL members and arbitrarily uploads malicious data to destroy global model



Impact of Attacks

- **Confidentiality:** Personal information such as medical records, Personally-identifiable information (social security, location, etc.), and payment details may be accessed through these attacks -> costly, privacy violation, unsafe
- **Integrity:** Leads to poor model performance -> waste of time and resources; Can lead to a specific model fault, such as in autonomous vehicle systems or a medical model incorrectly predicting a disease
- **Availability:** Unreliability of nodes in the system that may go undetected





03

Privacy Defenses

Defense of Attacks/Security Definitions

Privacy Defenses

Concern	Defense
Unauthorized Viewing	Encryption
Unauthorized Modification	Integrity Constraints
Untrusted Central Server	Differential Privacy
Untrusted/Byzantine Clients	Robust aggregation/Outlier Detection

Privacy Defenses

- **Differential privacy**
 - Limits the information revealed about a single entry in a database by adding noise or using regularization techniques, either locally or centrally.
- **Robust aggregation**
 - Detects and rejects corrupted updates from a fraction of clients to protect against network instabilities and malicious actors.
- **Outlier detection**
 - Examines clients' updates test error and discards those that will not improve the model's performance, protecting against malicious updates.

Security Definitions

Least Secure

Public

Fastest

Restricted

Sensitive

Most Secure

Secure

Slowest



Security Definitions

Security Level	Integrity	Encryption	Untrusted Server	Untrusted Clients	Federated Defenses
Public	✓	✗	✗	✗	✗
Restricted	✓	✓	✗	✗	✗
Sensitive	✓	✓	✓	✗	Differential privacy
Secure	✓	✓	✓	✓	Differential privacy, robust aggregation, and outlier detection

Future Work

- Choose concrete protocols for encryption and integrity constraints
- Pick specific implementations of differential privacy, robust aggregation, and outlier detection
- If time permits, develop a test implementation to demonstrate security levels and test time complexity



References

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THANKS!

Do you have any questions?

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